# AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

## THE SITE

| | |
|---|---|
| Location | CyrusOne CHI3, IL -- station KDPA, America/Chicago |
| Building | OSM 1481581947<br>CyrusOne CHI3 |
| Facade gap | not applicable -- one building, no second facade |
| Weather record | 43,457 real hourly records from KDPA |
| Plume physics | NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project. |

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

## THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

| | |
|---|---|
| Day | 2021-05-01 (crossing) |
| Plant limit | 21.0 C |
| Notice required | 3 h |
| Forecast skill | 0.50 relative to persistence |
| Level anchor | sensor |
| Condenser bank | longest facade |
| Switch budget | 2 mode changes per day |
| Minimum dwell | 3 h |
| Max dew point | 15.0 C (Green Grid WP#46 p.6) |

## WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 12 of 24 hours with 2 mode changes. The reactive incumbent took 14 hours -- 2 more -- but declared 2 unsafe hours against the agent's 0.

| | |
|---|---|
| Agent | 12 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s) |
| Incumbent | 14 of 24 hours free cooling, 2 mode change(s), 2 unsafe hour(s) |

## EVERY HOUR, AND THE REASON FOR IT

| hh | mode | safe | binding | bound | limit | actual | margin |
|---|---|---|---|---|---|---|---|
| 00 | FREE | yes | - | 4.715 | 21.0 | 4.440 | 1.002 |
| 01 | FREE | yes | - | 5.000 | 21.0 | 5.000 | 0.915 |
| 02 | FREE | yes | - | 4.165 | 21.0 | 3.330 | 0.842 |
| 03 | FREE | yes | - | 5.555 | 21.0 | 6.110 | 0.768 |
| 04 | FREE | yes | - | 6.390 | 21.0 | 7.220 | 0.717 |
| 05 | FREE | yes | - | 6.110 | 21.0 | 7.780 | 0.906 |
| 06 | FREE | yes | - | 8.080 | 21.0 | 8.330 | 0.892 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 07 | FREE | yes | - | 10.835 | 21.0 | 10.000 | 1.540 |
| 08 | FREE | yes | - | 13.610 | 21.0 | 12.220 | 2.078 |
| 09 | FREE | yes | - | 15.835 | 21.0 | 15.000 | 2.111 |
| 10 | FREE | yes | - | 17.225 | 21.0 | 16.670 | 1.831 |
| 11 | FREE | yes | - | 18.860 | 21.0 | 18.890 | 1.545 |
| 12 | mechanical | no | dry-bulb | 21.110 | 21.0 | 22.220 | 1.111 |
| 13 | mechanical | no | dry-bulb | 22.780 | 21.0 | 25.000 | 0.935 |
| 14 | mechanical | no | dry-bulb | 24.720 | 21.0 | 27.220 | 1.042 |
| 15 | mechanical | no | dry-bulb | 26.390 | 21.0 | 28.330 | 0.891 |
| 16 | mechanical | no | dry-bulb | 27.775 | 21.0 | 28.890 | 1.071 |
| 17 | mechanical | no | dry-bulb | 28.055 | 21.0 | 28.330 | 0.974 |
| 18 | mechanical | no | dry-bulb | 27.775 | 21.0 | 27.220 | 1.107 |
| 19 | mechanical | no | dry-bulb | 27.500 | 21.0 | 26.110 | 1.379 |
| 20 | mechanical | no | dry-bulb | 26.110 | 21.0 | 23.890 | 1.457 |
| 21 | mechanical | no | dry-bulb | 25.000 | 21.0 | 22.780 | 1.301 |
| 22 | mechanical | no | dry-bulb | 23.890 | 21.0 | 21.670 | 1.051 |
| 23 | mechanical | no | dry-bulb | 22.500 | 21.0 | 20.560 | 1.125 |

bound  = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

## THE REASONING, HOUR BY HOUR

**00:00   FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 4.715 C, which is 16.285 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.002 C, and the margin is measured, not chosen: 1.002 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

**01:00   FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 5.000 C, which is 16.000 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.915 C, and the margin is measured, not chosen: 0.915 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

**02:00   FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 4.165 C, which is 16.835 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.842 C, and the margin is measured, not chosen: 0.842 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

**03:00   FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 5.555 C, which is 15.445 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.768 C, and the margin is measured, not chosen: 0.768 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

**04:00   FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 6.390 C, which is 14.610 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.717 C, and the margin is measured, not chosen: 0.717 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

**05:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 6.110 C, which is 14.890 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.906 C, and the margin is measured, not chosen: 0.906 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

**06:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 8.080 C, which is 12.920 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.892 C, and the margin is measured, not chosen: 0.892 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

**07:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 10.835 C, which is 10.165 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.540 C, and the margin is measured, not chosen: 1.540 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

**08:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 13.610 C, which is 7.390 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 2.078 C, and the margin is measured, not chosen: 2.078 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

**09:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 15.835 C, which is 5.165 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 2.111 C, and the margin is measured, not chosen: 2.111 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

**10:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 17.225 C, which is 3.775 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.831 C, and the margin is measured, not chosen: 1.831 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

**11:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 18.860 C, which is 2.140 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.545 C, and the margin is measured, not chosen: 1.545 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

**12:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 21.110 C against a 21.0 C limit, so it fails by 0.110 C. It would take a limit 0.110 C higher, or a bound 0.110 C tighter, to change this hour.

**13:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 22.780 C against a 21.0 C limit, so it fails by 1.780 C. It would take a limit 1.780 C higher, or a bound 1.780 C tighter, to change this hour.

**14:00 MECHANICAL [dry-bulb]**
Mechanical. The upper bound on intake air is 24.720 C against a 21.0 C limit, so it fails by 3.720 C. It would take a limit 3.720 C higher, or a bound 3.720 C tighter, to change this hour.

**15:00 MECHANICAL [dry-bulb]**
Mechanical. The upper bound on intake air is 26.390 C against a 21.0 C limit, so it fails by 5.390 C. It would take a limit 5.390 C higher, or a bound 5.390 C tighter, to change this hour.

**16:00 MECHANICAL [dry-bulb]**
Mechanical. The upper bound on intake air is 27.775 C against a 21.0 C limit, so it fails by 6.775 C. It would take a limit 6.775 C higher, or a bound 6.775 C tighter, to change this hour.

**17:00 MECHANICAL [dry-bulb]**
Mechanical. The upper bound on intake air is 28.055 C against a 21.0 C limit, so it fails by 7.055 C. It would take a limit 7.055 C higher, or a bound 7.055 C tighter, to change this hour.

**18:00 MECHANICAL [dry-bulb]**
Mechanical. The upper bound on intake air is 27.775 C against a 21.0 C limit, so it fails by 6.775 C. It would take a limit 6.775 C higher, or a bound 6.775 C tighter, to change this hour.

**19:00 MECHANICAL [dry-bulb]**
Mechanical. The upper bound on intake air is 27.500 C against a 21.0 C limit, so it fails by 6.500 C. It would take a limit 6.500 C higher, or a bound 6.500 C tighter, to change this hour.

**20:00 MECHANICAL [dry-bulb]**
Mechanical. The upper bound on intake air is 26.110 C against a 21.0 C limit, so it fails by 5.110 C. It would take a limit 5.110 C higher, or a bound 5.110 C tighter, to change this hour.

**21:00 MECHANICAL [dry-bulb]**
Mechanical. The upper bound on intake air is 25.000 C against a 21.0 C limit, so it fails by 4.000 C. It would take a limit 4.000 C higher, or a bound 4.000 C tighter, to change this hour.

**22:00 MECHANICAL [dry-bulb]**
Mechanical. The upper bound on intake air is 23.890 C against a 21.0 C limit, so it fails by 2.890 C. It would take a limit 2.890 C higher, or a bound 2.890 C tighter, to change this hour.

**23:00 MECHANICAL [dry-bulb]**
Mechanical. The upper bound on intake air is 22.500 C against a 21.0 C limit, so it fails by 1.500 C. It would take a limit 1.500 C higher, or a bound 1.500 C tighter, to change this hour.

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